

Standard Operating Manual

Oxford Plasmalab 80 Plus Plasma Etcher

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1. Picture and Location

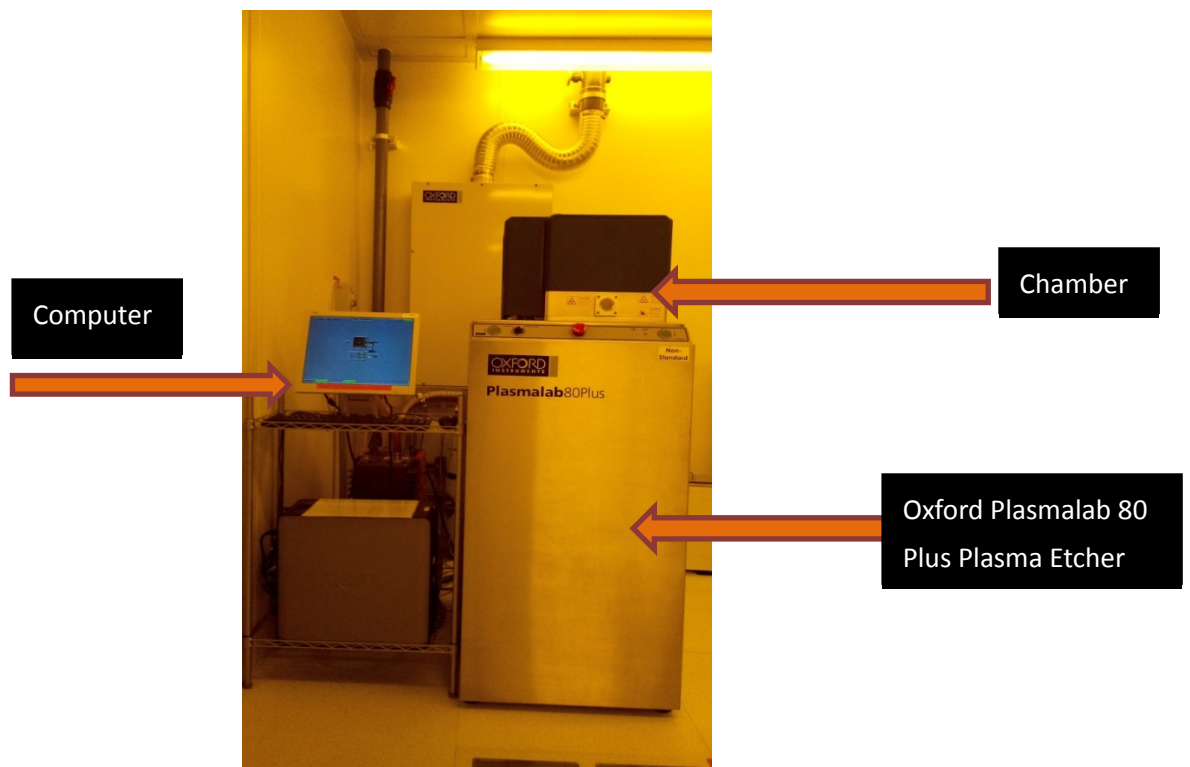


Fig 1: This tool is located at NFF Enterprise Center Cleanroom Room 4162

2. Process Capabilities

2.1 Cleanliness Standard

Oxford Plasmalab 80 Plus Plasma Etcher is “Non-Standard” equipment for dry etching process.

2.2 Available Etching Materials

The following sources are available in Oxford Plasmalab 80 Plus Plasma Etcher.

LTO, Thermo Oxide, Silicon Nitride, Poly-silicon and Silicon on 4" wafer or small piece sample.

NOTE: Consult NFF EC staff prior to do new materials other than above listed. **Hazardous** or **Radioactive** materials are not allowed to be etched in Oxford Plasmalab 80 Plus Plasma Etcher.

2.3 Performance of the Oxford Plasmalab 80 Plus Plasma Etcher

What CAN do:

- Oxford RIE 80+ can etch thin films thickness ($\geq 400\text{\AA}$ thick) or thick films thickness ($\leq 100\mu\text{m}$ thick).

What CANNOT do:

- Oxford Plasmalab 80 Plus Plasma Etcher could not replace ICP DRIE for etching thick films, e.g. over $100\mu\text{m}$
- It could not be used for photoresist Stripping, due to photoresist is for mask protection only.
- Users cannot modify the recipes' parameters, such as chamber pressure, RF power, process gas flow and chiller temperature.
- Process time cannot over 20 minutes per run and 5 -10 minutes s allowed for chamber cooling.

3. Contact List and How to Become a User

3.1 Emergency Responses and Communications

- Security Control Center: 2358-8999 (24hr) & 2358-6565 (24hr)
- Safety Officer: Mr. Wing Leong CHUNG 2358-7211 & 64406238
- Deputy Safety Officer: Mr. Man Wai LEE 2358-7900 & 9621-7708
- NFF EC Technician: Mr. Peter Yiu Cheong PUN 2358-7225 & 2358-7218
- NFF Phase 2 Technician: Mr. Wilson Pui Keung YIP 2358-7894

3.2 Training to Become a Qualified User

Please follow the procedure below to become a qualified user.

1. Read all materials on the NFF website concerning the Oxford RIE 80+.
2. Send an e-mail to NFF requesting Oxford RIE 80+ safety operation training.

Scheduling can take up to several weeks due to the many requests coming in for this tool.

4. Operating Procedures

4.1 System Description

Oxford Plasmalab 80 Plus Plasma Etcher is a plasma processing system, which can be configured to carry out reactive ion etching(RIE).

The chamber arrangement for a typical RIE process is shown as below:

The electrode is powered by a 13.56 MHz RF generator. The associated auto-matching tuner unit is positioned close to the driven electrode.

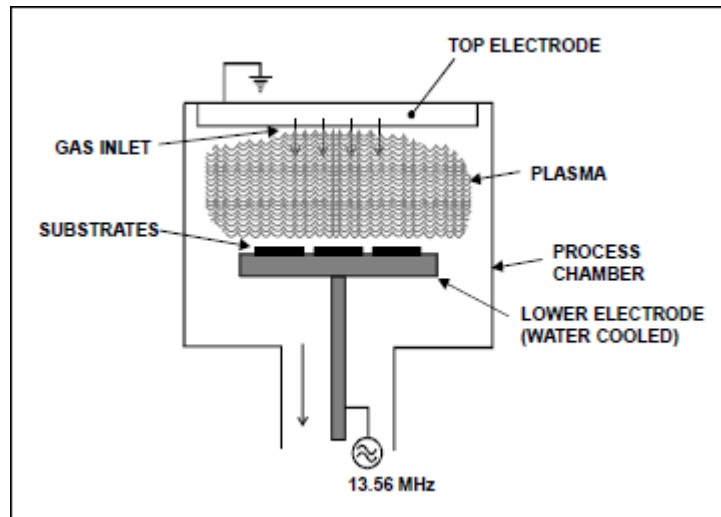


Fig 2: Typical RIE configuration

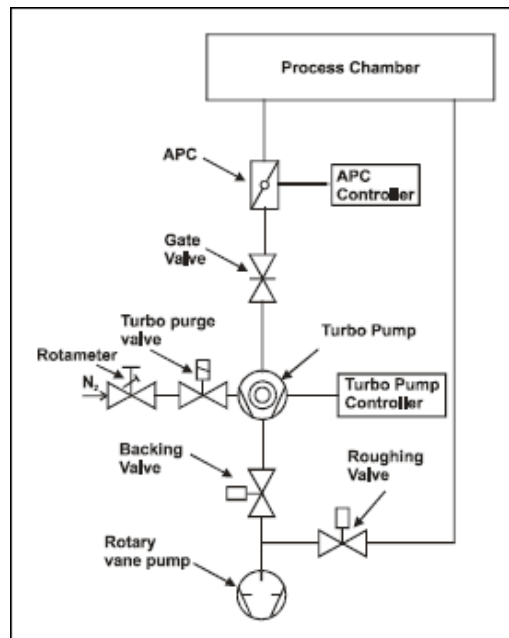


Fig 3: Backing and turbo molecular, turbo process and turbo bypass pump down

Backing and turbo pumps: process through the turbo pump and pump down by bypassing the turbo

With this arrangement, the chamber can be vented and roughed out without stopping the turbo pump.

Description of Oxford RIE 80+

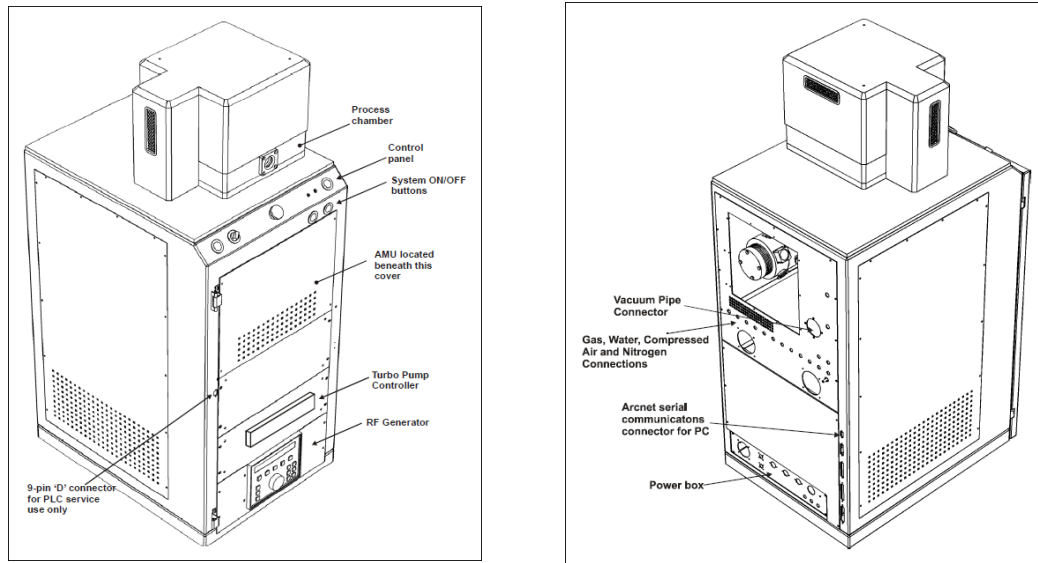


Fig 3: Description of major system components for Oxford RIE 80+.

A stainless steel cabinet with removable access panels encloses the mechanical and electronic components of the system and provides a support for the processing chamber. Turbo molecular and rough pumps are mounted separately from the main console cabinet.

The chamber lid and integral top electrode are raised and rotated clear of the chamber base and substrate table by a pneumatic hoist mechanism. This provides access to the table for loading and removal of substrates. The hoist is, for safety consideration, operated by pressing two buttons simultaneously.

The cylindrical aluminum chamber has a view port in its upper section and three ports, for roughing, extraction and pressure gauges, in its base. Gas is supplied to the upper electrode “shower head” via channels in the two halves of the chamber. Moreover,

coolant cooling is provided for the table where Reactive Ion Etching process is used.

Controls and indicators

The controls and indicators are mounted on the front of the unit as shown as below:

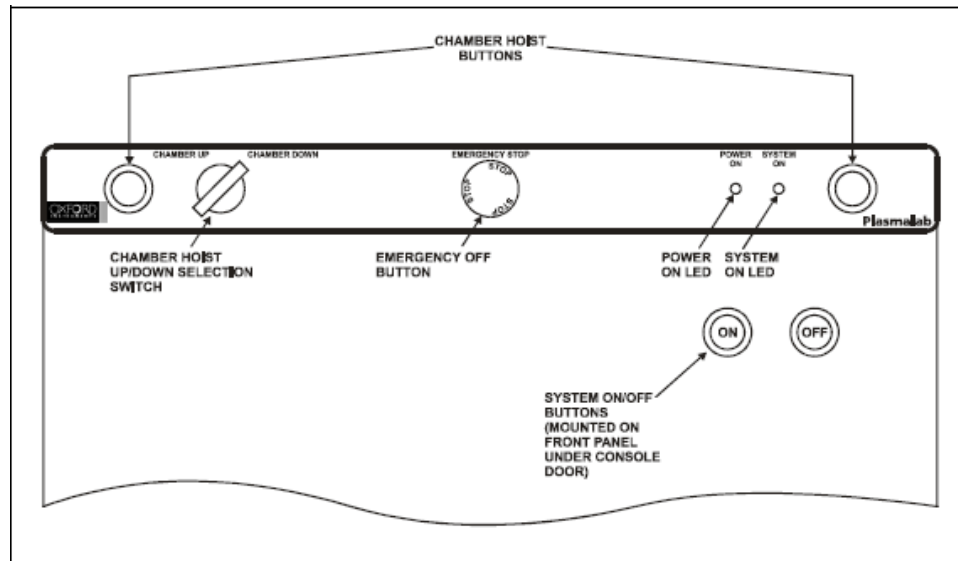


Fig 4: diagram of controls and indicators

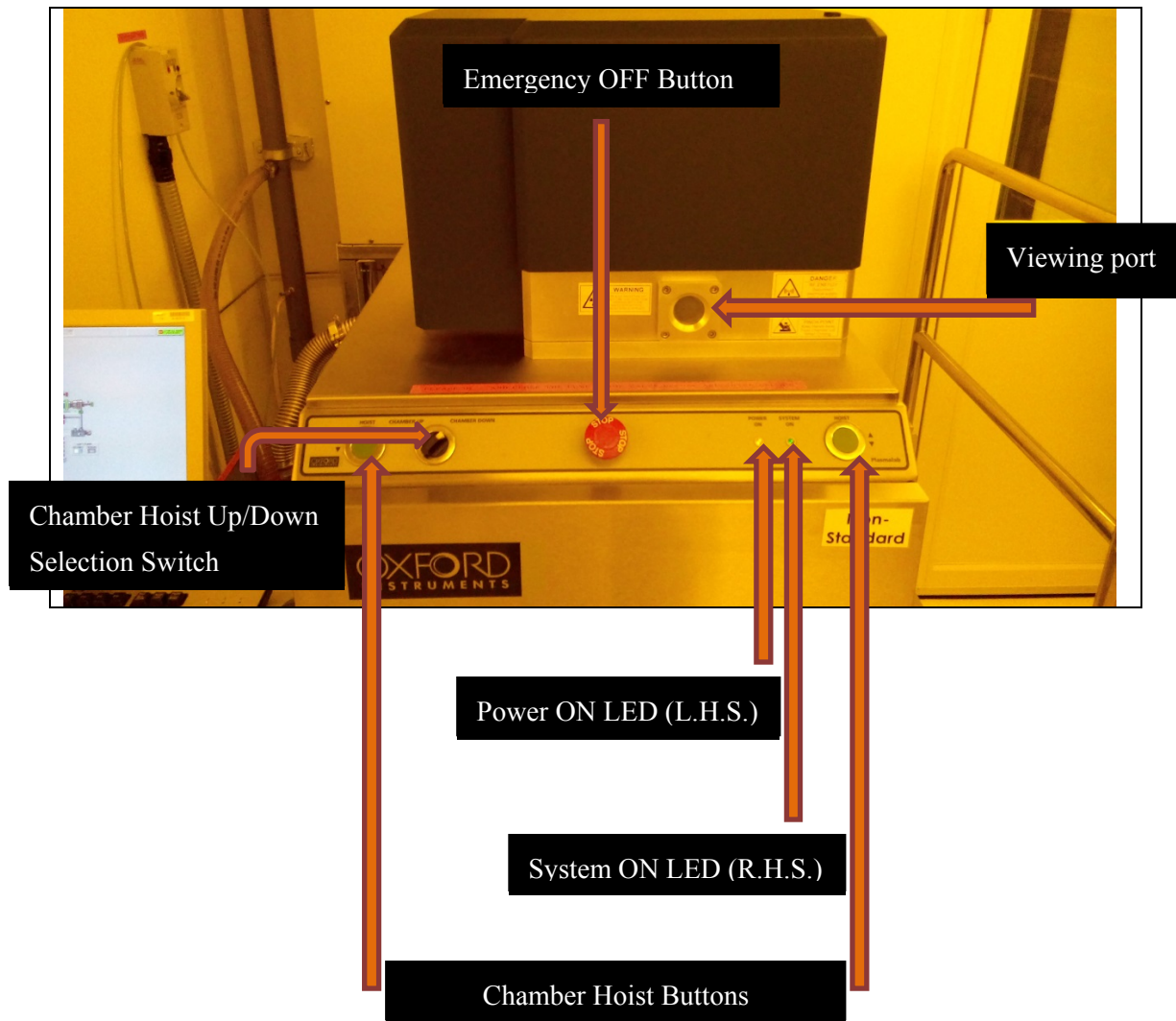


Fig 5: Real picture of of controls and indicators

Chamber Hoist buttons and up/down selection switch: To raise or lower the chamber top, the up/down selection switch is set to the required position, then both chamber hoist buttons are pressed.

Emergency Off (EMO) button: When pressed, the system is shut down.

Power On indicator: Indicates the electrical power is connected to the machine.

System On button: Switches the system power on.

System On indicator: Indicates that the system is powered up.

System Off button: Switches the system power off.

Emergency off and Interlock facilities

Emergency off (EMO) and interlock facilities are provided to shut down the machine in an emergency and to stop process until the system is fully initiated. An emergency off switch, with normally-closed contacts, is mounted on top of the console. The switch is activated by pressing a red “Emergency Off” button.

PLC interlock chain

The interlock chain is monitored by the software, but acts independently. It is also supplemented by machine protection sensors, which operate only via the software.

To enable RF power:

1. The 600 mbar vacuum switch (‘Vacstat’) must be at low pressure
2. The process chamber lid must be closed (or its hoist down)
3. The primary process pump must be running
4. The primary process pressure gauge (normally a capacitance manometer) must be on scale
5. The load lock inter-chamber valve (where fitted) must be closed
6. Customer-supplied external alarm devices must be at safe state
7. The inert gas purge to the primary process pump must be flowing.

To enable process gases:

1. RF power must be enabled.
2. The gas box lid must be closed.
3. Specific gases can be set in the gas box hardware to be mutually exclusive, so they cannot be turned on together.

Machine protections fitted where appropriate:

1. A nitrogen pressure switch, to detect adequate purge pressure to turbo molecular pump bearings or;
2. A nitrogen flow meter, to detect purge gas flow to pump bearings.
3. Water flow switch.

4.2 Safety Warnings

This equipment can cause injury if not used in a cautious manner.

1. Do not operate the system if any of the doors, panels or covers is removed.
2. Ensure that all personnel who operate this equipment are trained to use the equipment, and are alerted to the range of hazards present.
3. When opening the processing chamber, ensure that personnel stand clear of the chamber lid and hoist assembly.
4. Close chamber door carefully, ensure that personnel vacate the vicinity of the door and its operating mechanism before it is closed to avoid trapped fingers etc.
5. Ensure that you do not leave tweezers between the lid and the base, damage the equipment could occur.
6. When the compressed air is first applied to the system, the initial chamber lid movement will be repaid unless the “HOIST” buttons are operated intermittently.
7. If the sample to be etched contains photo resist, then the substrate must be baked immediately before placing it in the chamber.

8. Never smoke or eat in the “clean room” or where gases are stored. In addition to the fire risks and particulate contamination presented by smoking, some chemicals when burnt generate carcinogenic or toxic compounds.

Operation Rules

1. If an equipment failure while being used, never try to fix the problem by yourself. Please contact NFF staff.
2. Do not operate equipment unless you are properly trained and approved by NFF staff.
3. Do not leave an on-going experiment unattended.

4.3 Initial system checks

1. Make sure the System Interlocks (-15Volts, +15Volts, +24Volts, Gas-Pod Interlock and Water Flow Generator) are all at green light (ON Mode).
2. Make sure the Pump-down System (rough pump, APC valve, Gate valve, Turbo molecular pump (at speed) and Turbo backing valve) are all at green light (ON mode).
3. Make sure the Process Interlock are all shown ready, e.g. Lid->Close, Process Interlock->OK, Penning $1.88\text{e-}06\text{Torr}$ ->Base pressure reached ($1.80\text{e-}05\text{Torr}$), Cm gauge->0 mTorr and Vent Time left->0 secs.

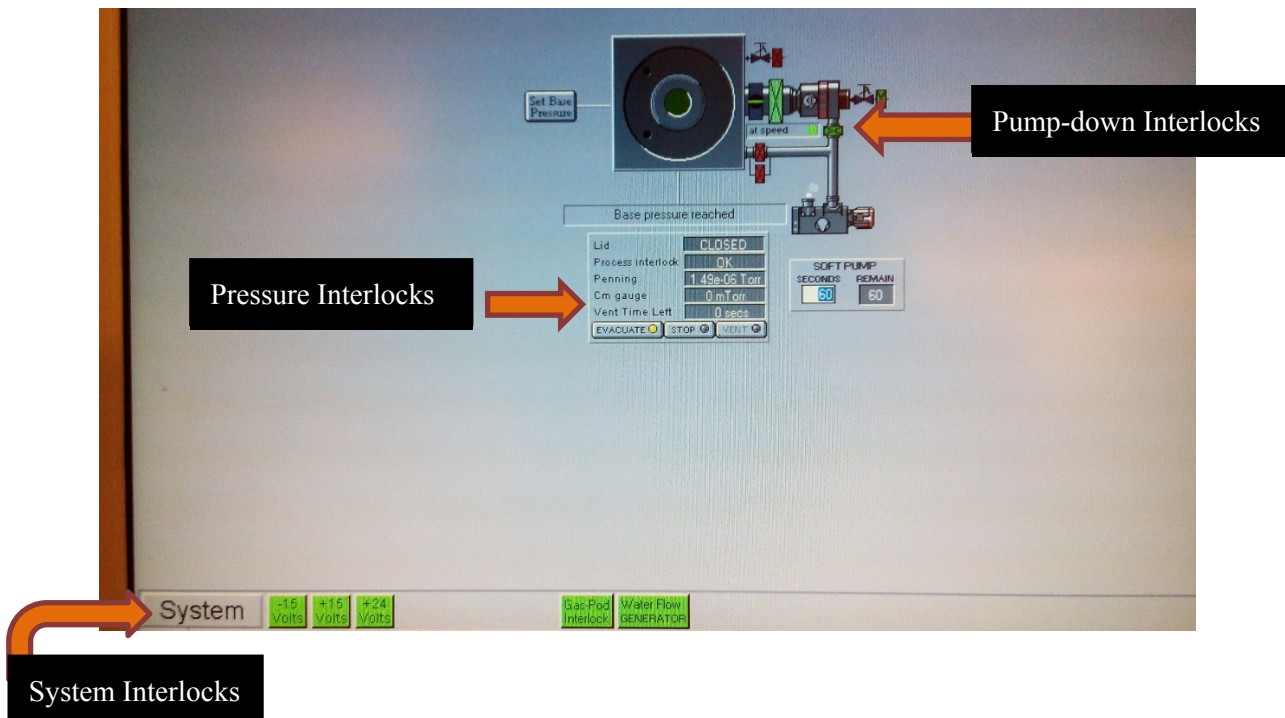


Fig 6: System, Process and Pump-down Interlocks

4.4 Status checks

1. Check the NFF website for reservations, problems and to see if it is already enabled by another user.
2. Check for an EMPTY sign attached to the machine. Do not use if an IN USE signs or MAINTENANCE sign is there. Check for problem notes.
3. The system is available if the initial system and status checks are normal. Check-in the equipment and enable the system on NFF Machine Reservation System. Place sign "IN USE" on the machine.

4.5 Venting the Chamber (Before Loading Wafers)

1. Ensure the Pump Control page is displayed (if necessary, select the System Menu, then the Pumping option).
2. Select the **STOP** button, then the **VENT** button for the process chamber. Note

that the vent sequence is controlled by a timer to allow time for the turbo pumps to be purged.

3. When the 'Vent Time Left' timer has decremented to zero, the process chamber has been vented. A clear gap between the chamber lid and the chamber should be seen.

4.6 Opening the Process Chamber

To open the process chamber when it has fully vented, see the following procedure:

1. At the control panel, set the up/down selection switch to its chamber up position.
2. Press both hoist buttons simultaneously. The chamber lid will raise and rotate.
3. When the chamber lid fully raised and stop rotate, release both hoist buttons.
4. If the lid does not open within 5 seconds, release the hoist buttons and try again. If the lid does not open after the second attempt, please contact NFF EC staff.

4.7 Inspecting the chamber before Dry Etching

1. Review the previous run in logbook to ensure there are no reported problems.
2. Fill in the log sheet with your name, NFF project number, reservation timeslot, chamber pressure, RF power, gases flow rate and processing time being used.
3. Keep out of moisture of the chamber by minimizing the time of it at ambient room temperature and pressure.

4.8 Load wafers

1. Place your sample face up in the center of the substrate holder. Before start up, please wear gloves to reduce contamination and protect the hands.
2. Maximum 3 pieces of 4” full wafers are allowed into the chamber per run.
3. Check the O-rings in the chamber lid and substrate holder to ensure that they are free of particles, dirt and no damage.

4.9 Closing the Process Chamber

To close the process chamber, follow the procedure as below:

1. At the control panel, set the up/down selection switch to its Chamber Down position.
2. Press both hoist buttons simultaneously. The chamber lid will lower and rotate.
3. When the chamber lid fully lowered and stop rotate, release both hoist buttons.
4. If the lid does not open within 5 seconds, release the hoist buttons and try again. If the second attempt fails, please contact NFF EC staff.

4.10 Pumping Down the System for Dry Etching

To pump the system down, follow this procedure:

1. Ensure the Pump Control page is displayed (if necessary, select the System Menu, then the Pumping option)
2. Ensure the status indicators (PSU Monitor, Water and Gas Pod Interlock) are in green.

3. Ensure the process chamber lid is in its “down” position. If you intend to carry out process run, you will need to open the process chamber lid, place a wafer on the wafer holder in the process chamber, then close the process chamber lid.
4. Select the **Set Base Pressure** button, and then enter the required process chamber base pressure if different from the default (1.80e-05Torr). Click OK.
5. Click on the rough pump mimic to start the pump.
6. Select the **Evacuate** button for the process chamber. You will be prompted to enter a wafer identity, either enter the identity and click **OK**, or click **Cancel** (to pump down without the wafer identity in the chamber). The relevant valves will operate and the process chamber will pumped down be automatically.

4.11 Start your Dry Etching Process

Automatic Process Run:

1. Pump the system down.
2. Check that the system has pumped down to the base pressure. (The process chamber message panel should display "Base Pressure reached").
3. Select the Process Menu, then the Recipe option. The recipe page is displayed.
4. Select the **Load** button and then select the required recipe from the displayed list. Click **OK**.
5. If the material you want to etch and/or the recipe is not in the system, please contact NFF EC staff and he will work with you to find an appropriate recipe from the system.

6. To run the loaded recipe, select the **Run** button. The recipe will be automatically run and the Chamber 1 page will be displayed to allow you to monitor its progress.
7. If there is anything wrong, press the **ABORT** button to abort the process. Then go to Step 4 and select the correct process.
8. Always verify by watching the view port to make sure the plasma is on and the brightness and colour is correct. (IMPORTANT: Do not stare through the viewing port at the plasma prolonged, viewing may damage your eyes)

Remarks:

You can pause the process at any time by selecting the **PAUSE** button. This will cause the Step Time and the plasma power to stop with the current step time indicated.

Re-starting the process will cause the process to continue from the time it was paused. If, during the pause period, you change any of the process parameters, e.g. gas demand, pressure etc.

You must press the **START** button for the changes made to come into effect, this will cause the step timer to continue from the time it was paused.

You can stop the process at any time by selecting the **STOP** button. The system will display the popup message "Process Complete", if required, you can then run the same or another process.

When the "Process Complete" popup message displays, the system can be vented

or another process run can be carried out.

Creating and Editing Recipes

This page is used to assemble and store in memory for all set points and instructions, which make up a Recipe for an Automatic Mode run. These recipes consist of a sequence of process Steps.

The **Recipe** option (accessed from the **PROCESS** menu) displays the recipe page for the process chamber. This page allows you to create / edit recipes and the recipe steps that they contain.

Note: Before creating / editing recipes, make sure that you understand the operation of key components of the system to ensure that recipes proceed as expected.

Working with Recipes

Recipes are "built" using existing recipe steps, and edited as required. Within a recipe, steps can be manipulated using the **Step Commands** pop-up menu (accessed by clicking on the Recipe Steps field).

■ **Edit Step**: Enables the selected (highlighted) step to be edited.

■ **Repeat Step**: Repeats all subsequent steps until a Loop Step is reached. This group of steps can be repeated any number of times. (When you select this option, you are prompted to enter the number of times the group of steps is to be repeated.)

■ **Loop Step:** Terminates a Repeat Step group

■ **Insert Step:** Creates "a gap"; above the selected step to allow another step to be dragged into the list.

■ **Delete Step:** Deletes the selected step from the list.

■ **Cancel:** Closes the **Step Commands** pop-up menu.

4.12 Venting the Chamber (Before Unloading Wafers)

1. Ensure that the Pump Control page is displayed (if necessary, select the System Menu, then the Pumping option).
2. Select the **STOP** button, then the **VENT** button for the process chamber. Note that the vent sequence is controlled by a timer to allow time for the turbo pumps to be purged.
3. When the 'Vent Time Left' timer has decremented to zero, the process chamber has been vented. A clear gap between the chamber lid and the chamber will be present.

4.13 Pumping the System Down for Idle

To pump the system down, follow the procedure as below:

1. When finish samples unloading, close the process chamber lid and ensure that the process chamber lid is in its "down" position.
2. Ensure that the Pump Control page is displayed (if necessary, select the System Menu, then the Pumping option)
3. Ensure that the status indicators (PSU Monitor, Water and Gas Pod Interlock)

are in green.

4. Select the **Set Base Pressure** button, and then enter the required process chamber base pressure if different from the default (1.80×10^{-5} Torr). Click OK.
5. Click on the rough pump mimic to start the pump.
6. Select the **Evacuate** button for the process chamber. You will be prompted to enter a wafer identity, please click **Cancel** to pump down without the wafer identity. The relevant valves will operate and the process chamber will be automatically pumped down.
7. DON'T GO AWAY YET!
8. Monitor the chamber pressure to the base pressure at 1.80×10^{-5} Torr.
9. Clean up the area and return items to the proper locations.
10. Place an **EMPTY** sign on the machine.
11. Fill in the logbook completely and write down any problems.

Appendix (Details information for recipes)

Table 1: Process Gases

Gas Number	Gas	Maximum Flow (sccm)
Gas 1	O ₂	100
Gas 2	Ar	100
Gas 3	CF ₄	100
Gas 4	CHF ₃	100
Gas 5	SF ₆	100
Gas 6	N ₂	100
Gas 7	He	100
Gas 8	H ₂	100

Table 2: Recipe Parameters for dry etching

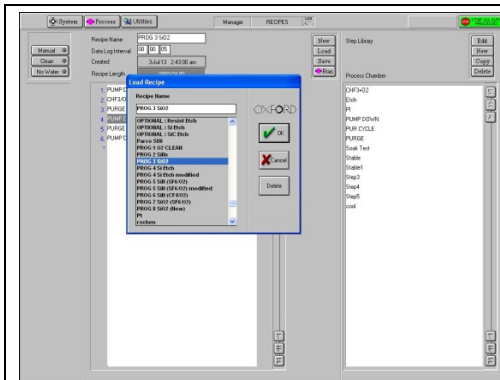
Recipe	Etch Materials	Gases and Flow (sccm)	Chamber Pressure (mTorr)	RF Power (Watts)
1	O ₂ CLEAN	80sccm O ₂	100	100
2	SiNx	5sccm O ₂ , 50sccm CHF ₃	55	100
3	SiO ₂	1.5sccm O ₂ , 48.5sccm CHF ₃	60	200
4	Si Etch	10sccm O ₂ , 50sccm SF ₆	150	100

Table 3: Recipe Parameters for dry etching

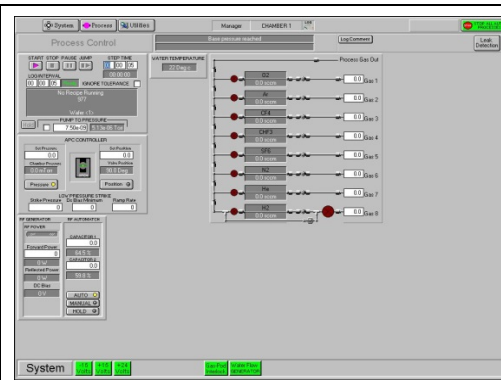
Recipe	Etch Materials	Step Function	Steps
1	O ₂ CLEAN	1. PUMP DOWN	1.80e-06 Torr
		2. O ₂ CLEAN	0 H *8 M 0 S
2	SiNx	1. PUMP DOWN	1.80e-06 Torr
		2. CHF ₃ /O ₂ (ER=1000A/MIN)	0 H *5 M 0 S
		3. PURGE CYCLE	0 H 0 M 30 S
		4. PUMP DOWN	0 H 0 M 30 S
		5. PURGE CYCLE	0 H 0 M 30 S
		6. PUMP DOWN	0 H 0 M 30 S
3	SiO ₂	1. PUMP DOWN	1.80e-06 Torr
		2. CHF ₃ /O ₂ (ER=400A/MIN)	0 H *5 M 0 S
		3. PURGE CYCLE	0 H 0 M 30 S
		4. PUMP DOWN	0 H 0 M 30 S
		5. PURGE CYCLE	0 H 0 M 30 S
		6. PUMP DOWN	0 H 0 M 30 S
4	Si Etch	1. PUMP DOWN	1.80e-06 Torr
		2. SF ₆ /O ₂ (ER=5000A/MIN)	0 H *5 M 0 S
		3. PURGE CYCLE	0 H 0 M 30 S
		4. PUMP DOWN	0 H 0 M 30 S
		5. PURGE CYCLE	0 H 0 M 30 S
		6. PUMP DOWN	0 H 0 M 30 S

“ * ” Remark: There is the processing time for dry etching.

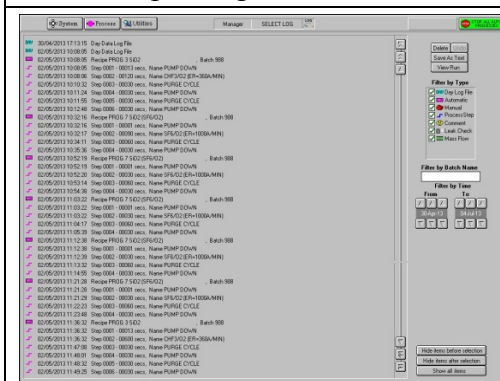
NANOSYSTEM FABRICATION FACILITY (NFF), HKUST



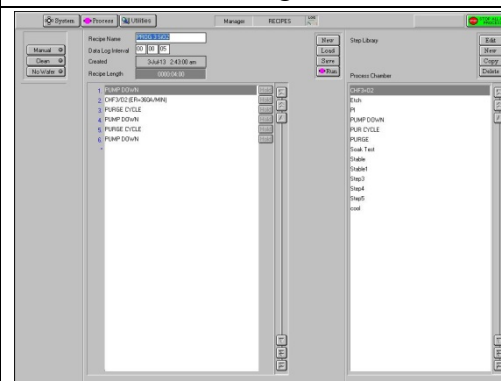
Load Recipes Page



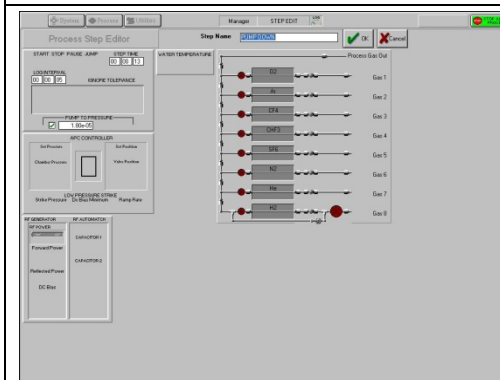
Process Chamber Page



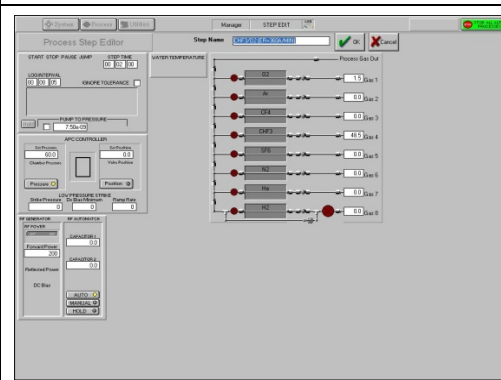
Process Log View Page



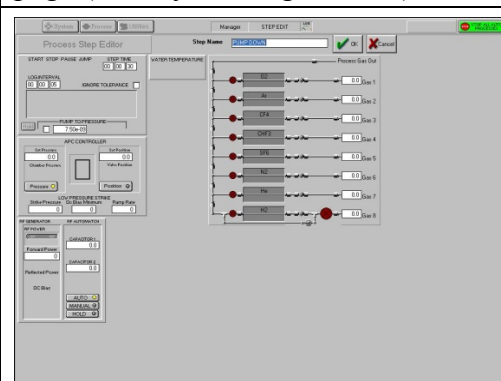
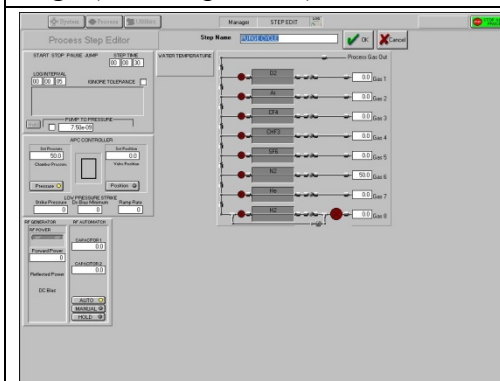
Process Recipes Page



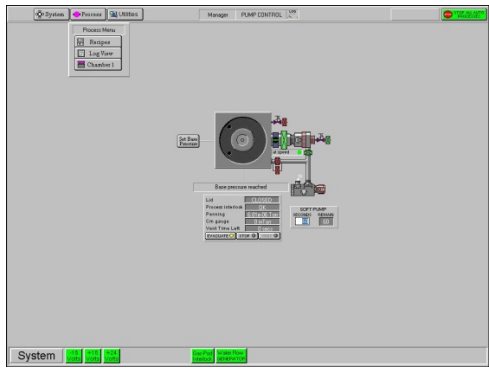
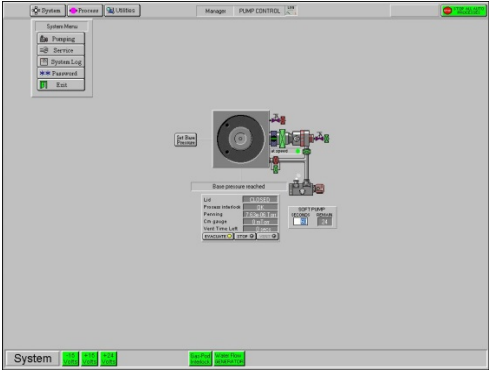
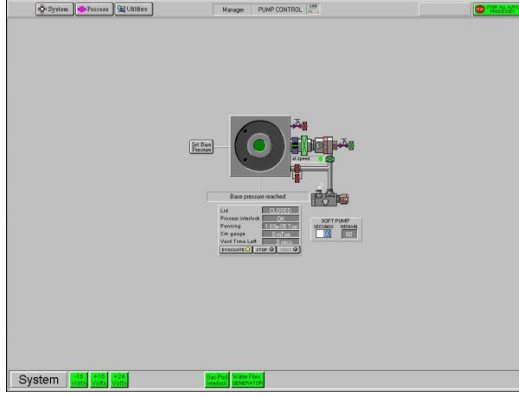
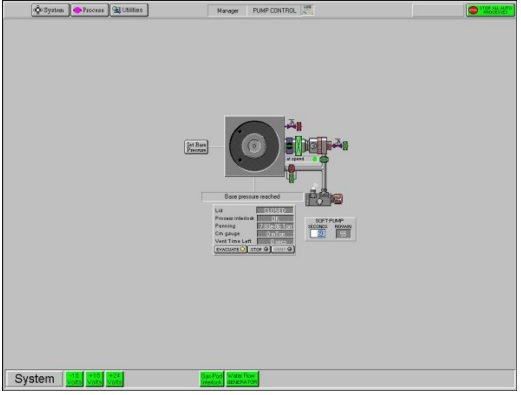
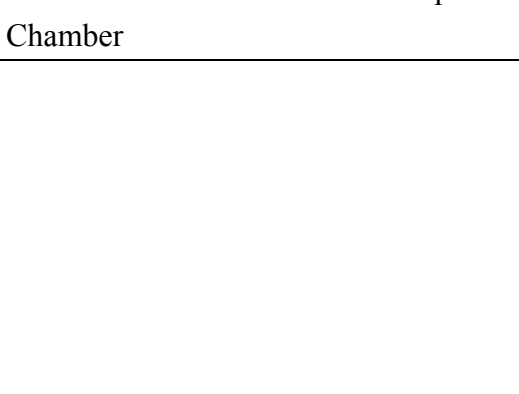
Process Recipes Process Step Editor Page (for Pump Down)



Process recipes Process Step Editor page (for Dry Etching Process)



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Process Recipes Process Step Editor Page (for N2 gas Purge Cycle) 	Process Recipes Process Step Editor Page (for Pump Down) 
Pull Down Menu - Process Menu 	Pull Down Menu - System Menu 
Chamber Status - With Sample in Chamber 	Chamber Status - Without Sample in Chamber 